



Acute Sleep Deprivation-Induced Anxiety and Disruption of Hypothalamic Cell Survival and Plasticity: A Mechanistic Study of Protection by Butanol Extract of *Tinospora cordifolia*

Payal Bajaj¹ · Gurcharan Kaur¹

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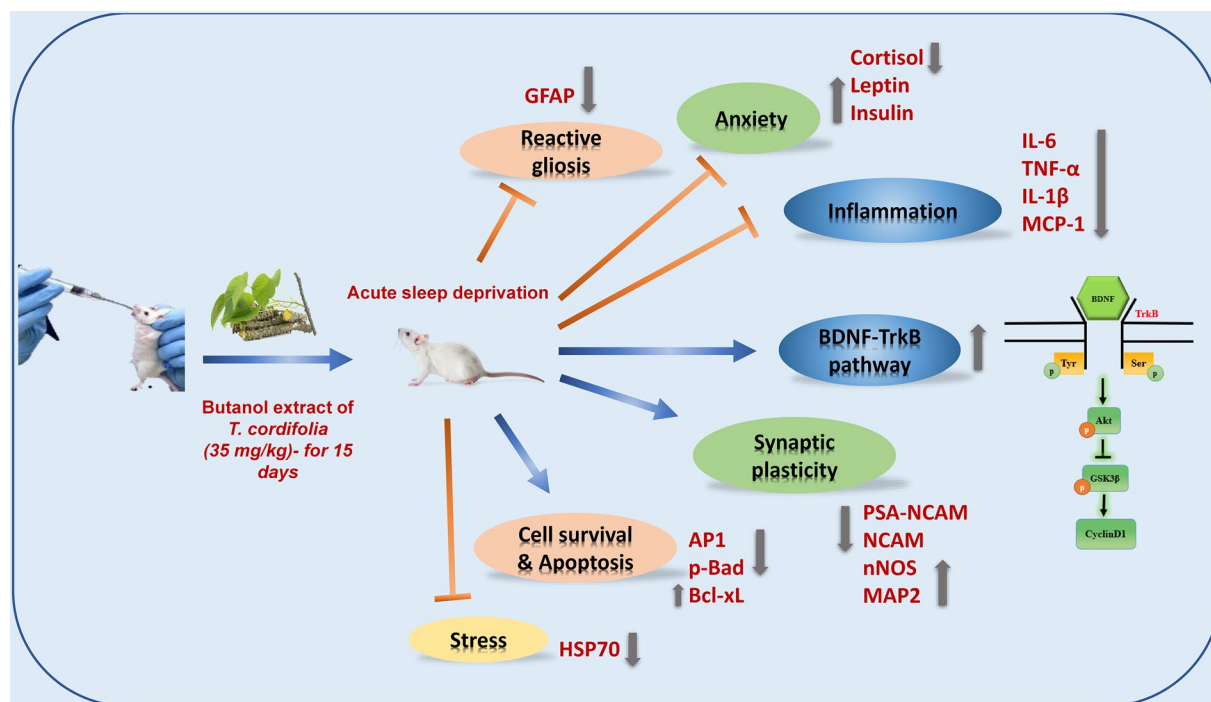
Abstract

Since sleep is a key homeostatic phenomenon of the body, therefore understanding the complex etiology of the neurological outcome of sleep deprivation (SD) such as anxiety, depression, cognitive dysfunctions, and their management is of utmost importance. The findings of the current study encompass the neurobehavioral as well as hormonal, and neuroinflammatory changes in serum and hypothalamus region of the brain as an outcome of acute SD and their amelioration by pre-treatment with butanol extract of *Tinospora cordifolia*. SD group animals showed anxiety-like behavior as evident from Elevated Plus Maze data and higher serum cortisol levels, whereas, pre-treatment with B-TCE showed anxiolytic activity and also reduced cortisol levels which was corroborated by an increase in leptin and insulin levels. Further, SD induced elevation of serum pro-inflammatory cytokines IL-6, TNF- α , IL-1 β , and MCP-1 and subsequent activation of astroglial cells in the hypothalamus was suppressed in B-TCE pre-treated animals. The current findings suggest that besides the cortical structures, hypothalamus region's synaptic plasticity and cell survival are adversely impacted by acute SD. Further active ingredients present in B-TCE may be useful for the management of SD-induced anxiety, systemic inflammation, and neuroinflammation by targeting hypothalamic BDNF-TrkB/PI3K-Akt pathways.

✉ Gurcharan Kaur
kgurcharan.neuro@yahoo.com

¹ Medical Biotechnology Laboratory, Department of Biotechnology, Guru Nanak Dev University, Amritsar, Punjab 143005, India

Graphical Abstract



Keywords *Tinospora cordifolia* · Anxiety · Hypothalamus · Inflammation · Neuroprotection · Synaptic plasticity

Introduction

Reduction in sleep quality and duration has become a global epidemic in modern society which may be attributed to unhealthy lifestyle choices, longer screen times, family or professional demands, some physical ailments or psychological issues, etc. Besides societal impact observed mainly in younger adults, aging is also associated with a decrease in average sleep duration along with disturbed sleep patterns. Over the last few decades, a marked decrease in average sleep time has been observed with a parallel increase in the prevalence of obesity and metabolic disorders, neuroendocrine dysfunctions, anxiety, cardiovascular diseases, etc. [1, 2]. SD is reported to cause oxidative damage, resulting in reactive oxygen species (ROS) formation which eventually leads to cellular and neuronal damage. The brain is more vulnerable to oxidative damage due to poor antioxidant machinery, abundant polyunsaturated fatty acids, high metabolic rate, and higher content of transition metals such as iron and copper [3, 4]. This oxidative damage impedes long-term potentiation (LTP), resulting in cognitive deficits such as depression, anxiety, and impaired learning and memory [5]. Further, SD as a metabolic challenge to the body is often associated with elevated levels of inflammatory

proteins which may be reflective of impaired physiological functions and initiation of the disease process [6, 7]. Several pre-clinical and clinical studies have established a bidirectional link between sleep deprivation and anxiety [8–10]. Although these studies have identified anxiety as a major neurobehavioral consequence of SD and explored its pathophysiology but the underlying molecular mechanisms still remain poorly understood.

The hypothalamus region of the brain regulates heterogeneous physiological functions such as neuroendocrine control, appetite, energy homeostasis, reproductive behavior, body temperature, movement initiation, etc. Previous studies exploring the effect of SD on the human and rat brain have largely focused on cortical structures leaving sub-cortical areas largely unexplored [11, 12]. Located at the base of the brain, the hypothalamus is considered a part of the limbic system and connects through reciprocal projections to the prefrontal cortex, hippocampus, and amygdala regions. Regulation of the sleep–wake cycle is one of the key functions of the hypothalamus [13] facilitated through hypothalamo-pituitary-adrenal (HPA) axis to trigger the release of the stress hormone, cortisol from the adrenal gland [14]. Further, hypercortisolism under stressful situations is closely associated with the development of mood and anxiety disorders [15]. Leptin, an

appetite-regulating hormone is also involved in cortisol axis regulation at the level of the hypothalamus [16, 17].

The neurotrophic hypothesis has recently gained much attention besides the involvement of monoamines in response to stressful conditions. BDNF and its cognate receptor TrkB are highly expressed in the hypothalamus and their expression is altered in response to stress as well as in inflammatory and neuropathic pain mechanisms [18]. Pharmacological interventions including benzodiazepines, non-benzodiazepines, and their analogs are widely used to alleviate the negative consequences of SD [19]. However, these drugs are associated with adverse side effects such as fatigue, disorientation, memory loss, and habituation [5]. Therefore, the use of traditional plant-based remedies owing to their neuroprotective activity may be a promising interventional approach to mitigate the adverse effects of SD on anxiety and cognition. *Tinospora cordifolia* also referred to as “Giloya” and “Guduchi” in the Indian traditional system of medicine is extensively used as a general tonic and re-juvenile herb in the treatment of different body ailments. It is available commercially either as a mono-therapy or as a component of polyherbal preparations [20, 21]. Extracts of this plant have been reported to possess several therapeutic properties including antioxidative [22], immunomodulatory [23], anti-inflammatory [24], antipyretic [25], and hepatoprotective [26] making it a promising source of natural medicines. Different extracts of this plant have been observed to show anti-inflammatory activity by modulating the expression of key inflammatory markers iNOS, pJNK, ICAM-1, COX-2 and pro-inflammatory cytokines level such as IL-6, IL-1 β , TNF- α in different disease models [27, 28]. A recent study has reported the neuroprotective activity of aqueous extract of *T. cordifolia* against MPTP-induced neurodegeneration and neuroinflammation (GFAP, Iba1, TNF- α , IL-1 β) and oxidative stress (SOD, GSH, CAT) markers were identified as molecular targets [29]. Additionally, oral administration of different extracts of *T. cordifolia* such as ethanolic, petroleum ether is also reported to exhibit neuroprotective activity bringing about improvement in motor coordination in animal models of depression and Parkinson's disease [30, 31].

Our lab has earlier reported that 50% ethanolic extract of *T. cordifolia* (TCE) possesses anxiolytic and neuroprotective activity against acute-sleep deprivation-induced anxiety and cognitive impairments in rats at a concentration of 140 mg/kg [32]. Butanol fraction of TCE (B-TCE) was further observed to possess neuroprotective and neuroregenerative activity at a much lower concentration (35 mg/kg) than the parent extract TCE against glutamate-mediated excitotoxicity under both in vitro [33] and in vivo [34] conditions. Recently, we have reported that B-TCE ameliorates cognitive and neuromuscular dysfunctions associated with acute SD and related signaling pathways were studied in the hippocampus and piriform cortex

regions of the brain [12]. Based on these previous lab findings, the current study was designed to understand the mechanistic aspects of B-TCE modulation of acute SD-induced stress and anxiety. Women, as they age, often suffer from sleep disturbances, inflammation, and adiposity which are attributed to a compromised immuno-neuro-endocrine axis mainly regulated at the hypothalamic level. The study used middle-aged female rats as an animal model as there is a paucity of data in literature in aged female rats. Gentle handling method was used to induce SD in rats and an elevated plus-maze test was performed on these animals to study their anxiety-like behavior. Investigation of the underlying mechanism(s) of anxiety-like behavior after acute SD was accomplished by targeting serum inflammatory markers and neuroinflammation, apoptosis, cell survival, and synaptic plasticity pathways in the hypothalamus region of the brain.

Materials and Methods

Collection, Identification, and Authentication of Plant Material

T. cordifolia stems climbing on Neem (*Azadirachta indica*) tree were obtained from a local forest located in the Ropar district of Punjab in the month of May. The specimen was authenticated and deposited at the herbarium of the Department of Botanical and Environmental Sciences, Guru Nanak Dev University, Amritsar, Punjab, India (Accession No. 65 Bot. & Env. Sc. dated 04-09 2017).

Preparation of B-TCE

B-TCE was prepared using the method previously described [33]. The stems of the plant were thoroughly washed with tap water and oven-dried at 45 °C. The stems were finely grounded (1 kg) and extracted with 50% ethanol using the percolation method. Excess solvent from the above-recovered extract was removed under reduced pressure using Rota-vapor (Buchi R-210, Switzerland) to yield 108 g of ethanolic extract (TCE). TCE so obtained was further fractionated in organic solvents (SRL, Analytical grade, 95% purity) in order of increasing polarity viz. hexane to yield hex-TCE fraction (0.245 g), Chloroform for Chl-TCE fraction (5.23 g), Ethyl acetate for Et-TCE fraction (3.833 g) and n-Butanol to recover B-TCE fraction (13.25 g). The fractions thus obtained were dried in a rotary evaporator.

Experimental Animals and Administration of the Extract

The experimental protocols were approved by the Institutional Animal Ethical Committee (IAEC) of Guru Nanak Dev University, Amritsar, Punjab, India (Reference no.:

226/CPCSEA/2019/10). All study procedures were performed in accordance with the IAEC guidelines for the care and use of animals. Female Wistar rats 13–15 months weighing 200–250 g were housed in polypropylene cages (3 rats per cage) at 25 ± 2 °C and 50% humidity. The animals were kept under a 12 h light/12 h dark cycle with an ad libitum access to food and water. After 1 week of acclimatization, rats were randomly divided into 4 groups of 6–7 rats each, namely Vehicle-undisturbed sleep (VUD), Vehicle-sleep deprived (VSD), B-TCE fed sleep-deprived (TSD), and B-TCE fed undisturbed sleep (TUD). VUD and VSD groups animals were given water as a vehicle, whereas, 35 mg/kg of B-TCE was given to TSD and TUD group animals between 9:00 and 10:00 am for 15 days using an oral gavage needle. B-TCE dose (35 mg/kg) used in this study was previously standardized in our lab [34]. On the 15th day, VSD and TSD group animals were sleep-deprived by the gentle handling method for 12 h in the light phase from 6 am to 6 pm, whereas, animals of the VUD and TUD group were left undisturbed. Whenever VSD and TSD group animals attempted to sleep, their cages were either shaken gently or animals were gently stroked with a soft brush. Post 12 h of SD, all the animals were exposed to the Elevated Plus Maze test.

Elevated Plus Maze Test

The elevated plus maze (EPM) was used to assess anxiety-like behavior and was performed immediately after 12 h of sleep deprivation. EPM set consists of two opposite open arms (50 cm × 10 cm) and two opposite closed arms (50 cm × 10 cm × 40 cm). The arms are connected through a central open area (10 cm × 10 cm), giving the apparatus an appearance of a plus sign. The apparatus is elevated 50 cm above the ground and was placed in a dimly lit room under noise-free conditions. Individual rat was placed in the center of the maze facing one of the open arms which was then allowed to explore the maze for 5 min. The ceiling-mounted video camera recorded the activity of each animal. Before placing the next animal, the apparatus was thoroughly wiped with 70% ethanol. Video recordings of each animal's performance were later analyzed by a blinded observer for the total time spent, number of entries, number of crossings in open arm, and head dip numbers. Entry into the open arm was counted only when all the four paws had crossed that arm. A crossing was scored when an animal entered into the open arm directly from its opposite arm. A head dip is characterized by the animal looking down from the open arm edge or central open area. Following EPM, animals were sacrificed and used for serum pro-inflammatory cytokines assay and protein markers expression in the hypothalamus region of the brain using western blotting.

Estimation of Pro-inflammatory Cytokines and Cortisol in Serum

Levels of IL-6, IL-1 β , TNF- α , MCP-1, insulin, and leptin in serum were measured using MILLIPLEX® MAP kit (Cat no- RADPKMAG-80K, Millipore, USA). According to the manufacturer's instructions, 200 μ L of assay buffer was added into each well of 96 well plate, and incubated for 10 min at 20–25 °C. Following incubation, assay buffer was completely removed and 25 μ L of standard samples were added into the designated wells while 25 μ L of assay buffer was added into the rest of the wells. 25 μ L of matrix solution was then added into all the wells followed by the addition of 25 μ L of serum sample into respective wells. 25 μ L of mixed beads were then added to all the wells and incubated overnight with gentle shaking at 4 °C. After three washings with 200 μ L of wash buffer, 50 μ L of detection antibodies were added and incubated for 2 h with gentle agitation, following which 50 μ L of Streptavidin–Phycoerythrin was added into wells and incubated for 30 min. The contents of the well were removed and the wells were then washed with 200 μ L of wash buffer. 100 μ L of sheath fluid was then added to each well with gentle agitation for 5 min and read on Luminex® software.

Cortisol was estimated using MILLIPLEX® MAP kit (Cat no- MSHMAG-21 K, Millipore, USA) following the manufacturer's instructions. After adding 25 μ L of standard and 25 μ L of serum samples in respective wells as discussed above, 25 μ L of HRP conjugate and 25 μ L of prepared beads were added to all the wells one after the other and incubated for 16–18 h at 4 °C. The contents were removed carefully followed by four times washing with 200 μ L of wash buffer. Later, 25 μ L of detection antibody was then added and incubated for 1 h at room temperature with gentle agitation and then proceeded as discussed above.

Western Blotting

Following EPM, animals (n = 6–7 per group) were anesthetized by thiopentone injection (1 unit per 10 g) and decapitated to dissect the brain. The hypothalamus region was dissected from the whole brain on ice and lysed in a lysis buffer (DTT, Protease inhibitor cocktail, Na₃VO₄, and 1× Tris Buffered Saline) using a tissue homogenizer. The homogenate was centrifuged at 8000 rpm for 15 min at 4 °C and the supernatant was collected. Total protein concentration was estimated in the supernatant with the Bradford method. Protein lysate (30–50 μ g) was resolved on an 8% and 10% SDS-PAGE and transferred onto a PVDF membrane with 0.45 μ m pore size (Hybond-P from Amersham Biosciences UK limited) using wet transfer apparatus (Bio-Rad, CA, USA). Membranes were then blocked with 5% skim milk for 2 h at room temperature, followed by incubation with respective

primary antibodies: mouse anti-GFAP (1:2500), mouse anti-Iba1 (1:1500), mouse anti-NCAM (1:1500), mouse anti-PSA-NCAM IgM (1:1000), mouse anti-HSP70 (1:5000), mouse anti-MAP2 (1:2000), rabbit anti-AP1 (1:1500), rabbit anti-p-Bad (1:1500), rabbit anti-TrkB (1:1500), rabbit anti-BDNF (1:1500), rabbit anti-p-Akt (1:2000), rabbit anti-pan-Akt (1:2500), mouse anti-cyclinD1 (1:3000) (All procured from Sigma-Aldrich, USA), mouse anti-Bcl-xL (1:2000), rabbit anti-p-GSK3 β (1:1000), rabbit anti-nNOS (1:1000) (All procured from Cell Signaling Technology, USA) for overnight at 4 °C. The membranes were also probed with mouse anti- α -tubulin (1:5000) (Sigma-Aldrich, USA) as an internal loading control. Membranes were then washed with 0.1% TBST thrice for 10 min each followed by incubation with respective HRP-labelled secondary antibodies (GeNei, India) at 25 °C for 2 h. Following incubation with secondary antibody, the membranes were washed with 0.1% TBST thrice for 10 min each. Finally, the blots were detected with Amersham ECL Plus Western blot detection kit and photographed using ImageQuant LAS 4000 (GE Healthcare, UK). Change in the protein expression was the average of at least three independent IDV values obtained by intensity analysis using AlphaEase FC software (Alpha Innotech CA, USA). The data is presented as the ratio of protein/ α -tubulin (except p-Akt) taking the value of VUD as 100%.

Statistical Analysis

The data was calculated as mean $\pm$ SEM and analyzed using GraphPad Prism Software (Version 8) from at least three independent experiments. Two-way ANOVA (Tukey's post hoc method) was used for comparison among different groups using sleep deprivation factor and B-TCE factor considering, 2 levels (Control vs SD) $\times$ 2 levels (Vehicle vs B-TCE). Values with $p \leq 0.05$ were considered to be statistically significant.

Results

B-TCE Suppressed Anxiety-Like Behavior in Sleep-Deprived Rats

EPM test was performed to assess the anxiety-like behavior of animals. Experimental rats were naïve to the study as they were not previously exposed to the apparatus to study their response to spatial novelty. Data analysis of the total time spent by animals in the open arms revealed a significant main effect of SD $\times$ B-TCE interactions ($F_{(1, 20)} = 4.629$, $p = 0.0438$) (Fig. 1a). Similarly, data for the total number of entries in open arms indicated a significant main effect of SD $\times$ B-TCE interactions ($F_{(1, 20)} = 5.632$, $p = 0.0278$) with post-hoc test suggesting a significantly fewer number of

entries ($p = 0.0257$) by VSD animals as compared to VUD group. This anxiety-like behavior was effectively prevented in B-TCE supplemented TSD animals which were observed to have significantly more number of entries ($p = 0.0426$) in the open arm as compared to VSD animals (Fig. 1b). Further, data analysis of the number of crossings suggested a significant main effect of SD $\times$ B-TCE interactions ($F_{(1, 20)} = 8.305$, $p = 0.0092$) (Fig. 1c). Although TSD group animals spent about 30% more time in the open arms with a 3X higher number of crossings but the change was not found to be statistically significant. The number of head dips which is suggestive of exploratory behavior in rodents was also analyzed which indicated a significant main effect of SD $\times$ B-TCE interactions ($F_{(1, 16)} = 22.88$, $p = 0.0002$) (Fig. 1d). Further, a post-hoc test revealed a significant reduction ($p = 0.0062$) in the number of head dips in the VSD group as compared to the VUD animals. On the contrary, TSD animals showed a significantly greater ($p = 0.0033$) number of head dips as compared to VSD animals, thus suggesting their exploratory behavior attributed to B-TCE (Fig. 1d).

B-TCE Modulated Blood Serum Cortisol Levels and Increased Leptin and Insulin Levels

Upregulation in the expression of serum cortisol levels was observed in response to sleep deprivation in the VSD group as compared to the VUD group animals which were normalized to control levels by B-TCE pre-treatment in both TSD and TUD groups, though the change was not statistically significant (Fig. 2a). Conversely, levels of serum leptin revealed significant main effect of SD $\times$ B-TCE interactions ($F_{(1, 16)} = 37.98$, $p < 0.0001$) with significant downregulation ($p = 0.0014$) in the VSD group as compared to VUD group which was restored significantly ($p = 0.0042$) by B-TCE supplementation (Fig. 2b). Further, levels of insulin revealed the significant main effect of SD ($F_{(1, 12)} = 5.355$, $p = 0.0392$) but no significant effect of B-TCE and SD $\times$ B-TCE interactions with post hoc test depicting significant downregulation ($p = 0.0448$) in VSD group as compared to the control VUD group and these changes were suppressed with prior supplementation with B-TCE in TSD and TUD groups (Fig. 2c).

B-TCE Modulated Inflammatory Response and Suppressed Reactive Gliosis

Since anxiety is inextricably linked with inflammatory response, so we further assayed the serum levels of pro-inflammatory cytokines. Both IL-6 and MCP-1 showed significant main effect of SD [IL-6: ($F_{(1, 20)} = 5.006$, $p = 0.0368$); MCP-1: ($F_{(1, 12)} = 4.837$, $p = 0.0482$). Though, the levels of both IL-6 and MCP-1 showed upregulation in the VSD group as compared to VUD animals which were ameliorated with B-TCE supplementation in the

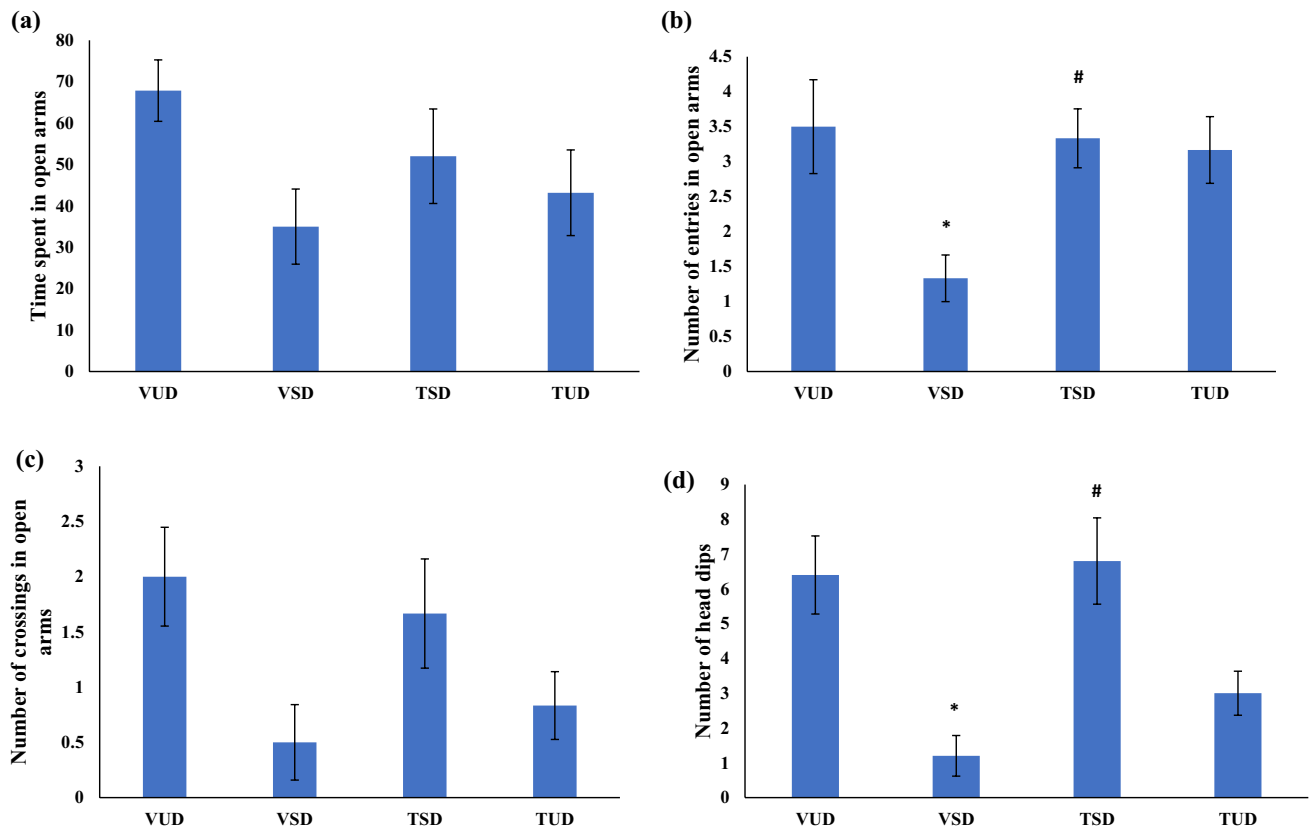


Fig. 1 B-TCE supplementation suppressed anxiety-like behavior in sleep-deprived rats ($n=6-7$ per group): An Elevated plus-maze experiment was performed to assess the anxiety level of rats by measuring **a** time spent **b** number of entries **c** number of crossings in the open arm. B-TCE supplemented TSD animals showed anxiolytic behavior as depicted by more time spent with entries and crossings in

the open arm as compared to VSD animals. **d** Histogram depicts the number of head dips by each group animal. Values are expressed as mean $\pm$ SEM. * $p \leq 0.05$ VUD versus VSD, TSD, TUD, # $p \leq 0.05$ VSD versus TSD and TUD, § $p < 0.05$ TSD versus TUD, Tukey's post hoc method after two-way ANOVA

TSD group, but the change was not statistically significant (Fig. 3a, d). TNF- α levels indicated significant main effect of SD $\times$ B-TCE interactions ($F_{(1,12)}=8.092$, $p=0.0148$) and SD ($F_{(1,12)}=9.972$, $p=0.0083$) with significant upregulation ($p=0.0054$) in VSD group as compared to the VUD group animals (Fig. 3b). Further, levels of IL-1 β revealed significant main effect of SD $\times$ B-TCE interactions ($F_{(1,16)}=17.62$, $p=0.0007$), B-TCE ($F_{(1,16)}=18.22$, $p=0.0006$) and SD ($F_{(1,16)}=15.06$, $p=0.0013$). Post-hoc test indicated significant upregulation ($p=0.0002$) in the VSD group as compared to the VUD group which was significantly downregulated ($p=0.0001$) by B-TCE supplementation in the TSD group animals (Fig. 3c). The expression of GFAP, a protein marker implicated in neuroinflammation was studied using western blotting in hypothalamus tissue lysate. Expression levels of GFAP showed significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)}=12.29$, $p=0.0080$), B-TCE ($F_{(1,8)}=14.48$, $p=0.0052$) and SD ($F_{(1,8)}=51.55$, $p<0.0001$) with significant upregulation ($p=0.0003$) in VSD group as compared to the VUD group. On pre-treatment with B-TCE, significantly

reduced ($p=0.0038$) expression of GFAP was observed in TSD animals as compared to the VSD group.

B-TCE Modulated the Expression of Synaptic Plasticity and Heat Shock Proteins

As the hypothalamus plays important role in plasticity and neurogenesis, therefore, the expression of key plasticity regulating proteins, PSA-NCAM and NCAM was studied in the hypothalamus region of the brain. Western blotting analysis of PSA-NCAM indicated a significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)}=8.079$, $p=0.0217$) and SD ($F_{(1,8)}=17.33$, $p=0.0032$) with significant upregulation ($p=0.0049$) in the hypothalamus region of VSD animals as compared to the VUD animals which was ameliorated by B-TCE supplementation in TSD and TUD group animals (Fig. 4a, b). Consistent with these results, analysis of NCAM expression also revealed the main effect of SD ($F_{(1,8)}=29.92$, $p=0.0006$) with significant upregulation ($p=0.0045$) in the VSD animals as

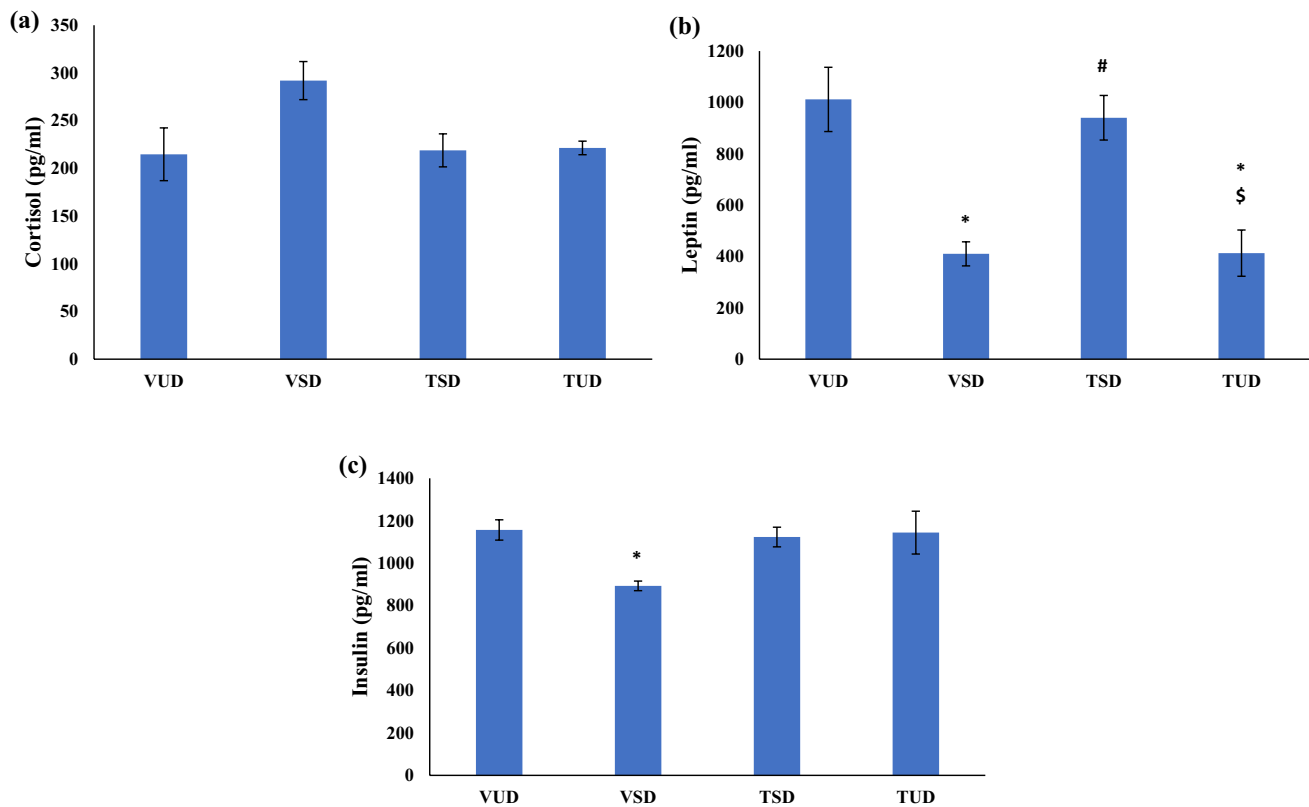


Fig. 2 B-TCE supplementation suppressed blood serum cortisol levels and increased leptin and insulin levels: **a–c** histograms represent the blood serum levels of cortisol, leptin, and insulin among four groups ($n=6-7$ per group). Values are expressed as mean $\pm$ SEM.

* $p \leq 0.05$ VUD versus VSD, TSD, TUD, # $p \leq 0.05$ VSD versus TSD and TUD, \$ $p < 0.05$ TSD versus TUD, Tukey's post hoc method after two-way ANOVA

compared to the VUD animals which was mitigated by B-TCE supplementation in TSD and TUD group animals (Fig. 4a, b). Further, expression of n-NOS indicated significant main effect of B-TCE ($F_{(1,8)} = 20.58$, $p = 0.0019$) and SD ($F_{(1,8)} = 11.95$, $p = 0.0086$) but no SD $\times$ B-TCE interactions. However significant downregulation ($p = 0.0178$) in VSD group was observed as compared to VUD group which was effectively restored ($p = 0.0066$) by B-TCE supplementation in TSD group animals. Analysis of MAP2 indicated significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)} = 37.83$, $p = 0.0003$), B-TCE ($F_{(1,8)} = 87.92$, $p < 0.0001$) and SD ($F_{(1,8)} = 11.28$, $p = 0.0099$) with significant downregulation ($p = 0.0007$) in VSD group in comparison to VUD group which was significantly ameliorated ($p < 0.0001$) by B-TCE pre-treatment in both TSD and TUD group animals. However, HSP70 expression analysis suggested the main effect of SD ($F_{(1,8)} = 6.942$, $p = 0.0300$) but with no significant change in expression in VSD group animals as compared to VUD group animals (Fig. 4a, b).

B-TCE Upregulated BDNF-TrkB and PI3K/Akt Pathway Implicated in Synaptic Plasticity and Cell Survival

To investigate the potential underlying molecular mechanism of neuroprotection by B-TCE the expression of downstream molecules of the BDNF-TrkB pathway was studied in the hypothalamus region after acute SD using western blotting. BDNF expression analysis indicated significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)} = 7.099$, $p = 0.0286$). TrkB expression analysis depicted significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)} = 60.95$, $p < 0.0001$) and B-TCE ($F_{(1,8)} = 78.38$, $p < 0.0001$) with significantly reduced ($p = 0.0009$) expression in VSD group animals as compared to VUD animals. B-TCE pre-treatment significantly ($p < 0.001$) restored the expression in TSD and TUD group animals (Fig. 5a, b). Further, p-Akt/pan-Akt expression analysis revealed significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)} = 12.39$, $p = 0.0078$), B-TCE ($F_{(1,8)} = 5.862$, $p = 0.0418$) and SD ($F_{(1,8)} = 5.458$, $p = 0.0477$). There was significant downregulation ($p = 0.0138$) in the VSD group in comparison to the

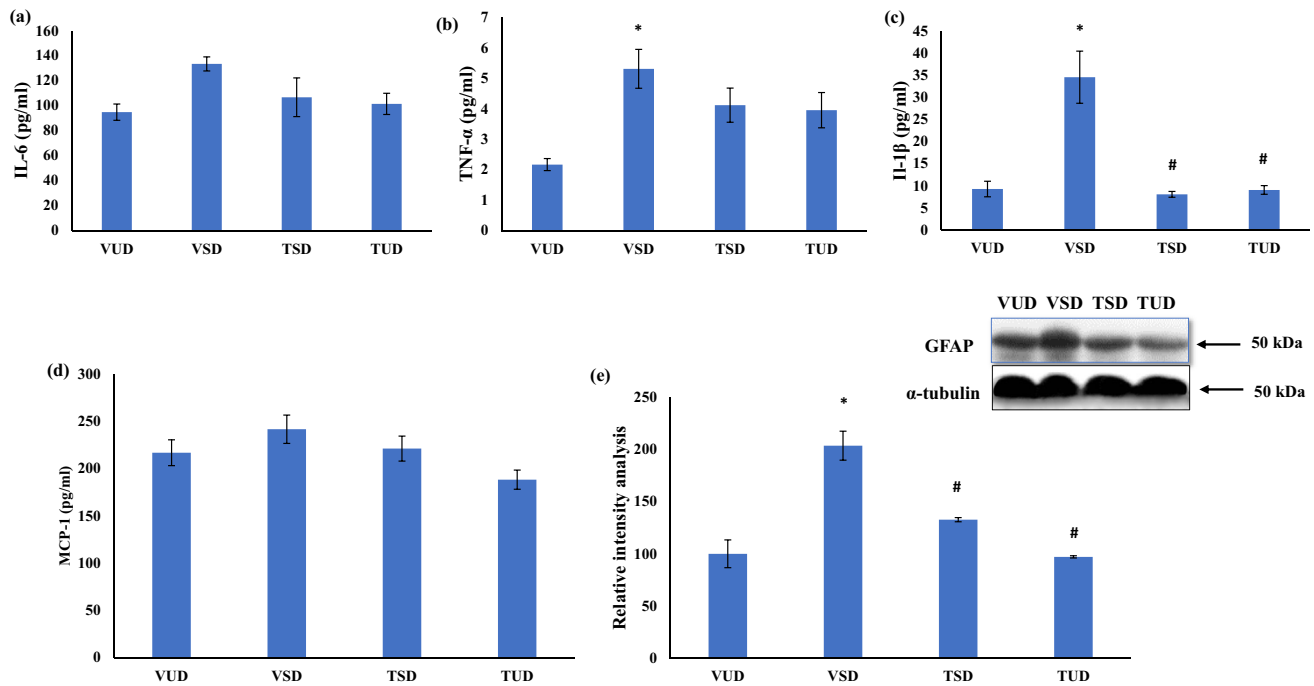


Fig. 3 B-TCE suppressed pro-inflammatory cytokines levels and reactive gliosis: **a–d** histograms represent the changes in the serum levels of IL-6, TNF-α, IL-1β, MCP-1. **e** The histogram represents the quantitative densitometric analysis of change in the expression of

GFAP among four groups ($n=6-7$ per group). Values are expressed as mean $\pm$ SEM. * $p \leq 0.05$ VUD versus VSD, TSD, TUD, # $p \leq 0.05$ VSD versus TSD and TUD, § $p < 0.05$ TSD versus TUD, Tukey's post hoc after two-way ANOVA

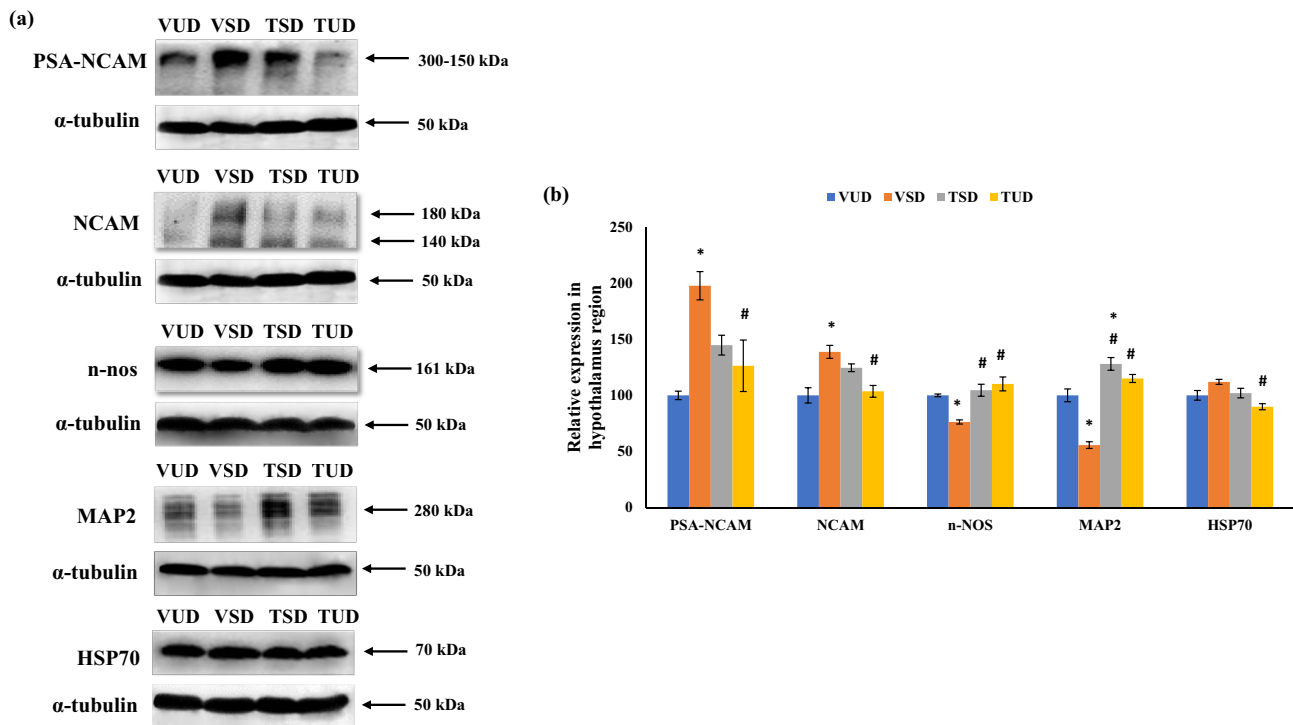


Fig. 4 B-TCE modulated the expression of synaptic plasticity and heat shock proteins: **a** representative western blot images of PSA-NCAM, NCAM, nNOS, MAP2, and HSP-70 in the hypothalamus region of VUD, VSD, TSD, and TUD rats ($n=6-7$ per group). **b, c**

Histograms representing their quantitative densitometric analysis among different groups are plotted as mean $\pm$ SEM. * $p \leq 0.05$ VUD versus VSD, TSD, TUD, # $p \leq 0.05$ VSD versus TSD and TUD, § $p < 0.05$ TSD versus TUD, Tukey's post hoc after two-way ANOVA

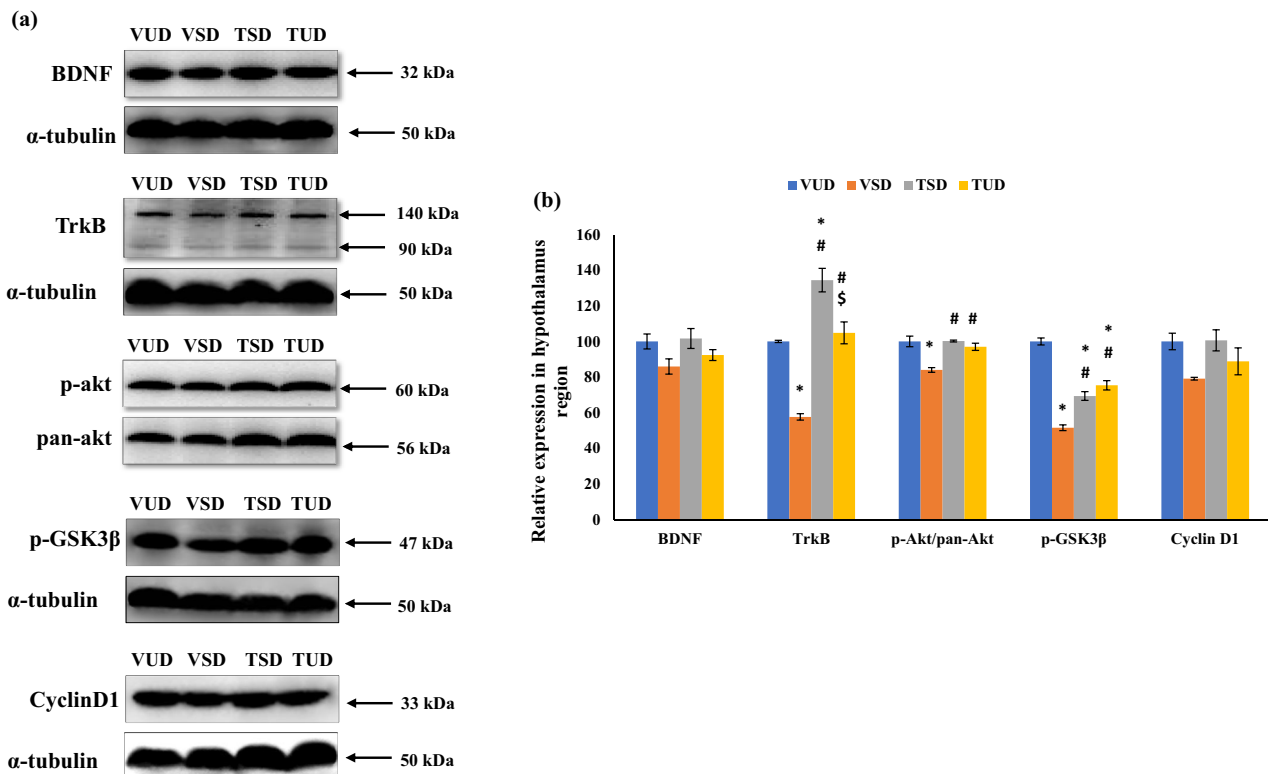


Fig. 5 B-TCE inhibited apoptosis: **a** representative western blot images of AP1, p-Bad, and Bcl-xL in the hypothalamus region of VUD, VSD, TSD, and TUD rats ($n=6-7$ per group). **b, c** Histograms representing their quantitative densitometric analysis among different

treatment groups are plotted as mean $\pm$ SEM. * $p \leq 0.05$ VUD versus VSD, TSD, TUD, # $p \leq 0.05$ VSD versus TSD and TUD, \$ $p < 0.05$ TSD versus TUD, Tukey's post hoc after two-way ANOVA

VUD group which was ameliorated ($p \leq 0.05$) by B-TCE pre-treatment in both TSD and TUD group animals. Similarly, p-GSK-3 β expression indicated significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)} = 92.60$, $p < 0.0001$) and SD ($F_{(1,8)} = 152.9$, $p < 0.0001$) with post-hoc test suggesting significant downregulation ($p < 0.0001$) of expression in VSD group which was restored ($p \leq 0.01$) by B-TCE supplementation. However, no significant change in the expression of BDNF and Cyclin D1 was observed in the VSD group animals as compared to control animals (Fig. 5a, b).

B-TCE Inhibited Apoptosis

To evaluate the effect of acute SD on cell apoptosis, the expression of key apoptosis regulating proteins AP1, p-bad, and Bcl-xL was studied. AP1 and p-bad expression analysis revealed significant main effect of SD $\times$ B-TCE interactions [AP1: ($F_{(1,8)} = 10.18$, $p = 0.0128$); p-bad: ($F_{(1,8)} = 43.29$, $p = 0.0002$)], B-TCE [AP1: ($F_{(1,8)} = 13.26$, $p = 0.0066$); p-bad ($F_{(1,8)} = 14.25$, $p = 0.0054$)], and SD [AP1: ($F_{(1,8)} = 20.92$, $p = 0.0018$); p-bad ($F_{(1,8)} = 148.9$, $p < 0.0001$)] with significant upregulation ($p \leq 0.01$) in the VSD group as compared to the VUD group animals. The

expression of these markers was alleviated ($p \leq 0.01$) by B-TCE supplementation in TSD and TUD group (Fig. 6a, b). Further, Bcl-xL expression revealed significant main effect of SD $\times$ B-TCE interactions ($F_{(1,8)} = 12.29$, $p = 0.0080$) and B-TCE ($F_{(1,8)} = 8.817$, $p = 0.0179$). Although no significant change in the expression of VSD group relative to VUD group was observed but a significant upregulation ($p = 0.0078$) in the expression of TSD group was observed as compared to the VSD group animals (Fig. 6a, b). These results, therefore, indicate that B-TCE could attenuate protein expression related to acute SD-induced apoptosis.

Discussion

The current pre-clinical study was aimed to investigate the role of the hypothalamus region in mediating psychosocial stress of acute SD and ensuing inflammation and anxiety-like behavior using middle-aged female rats as a model system as females are reported to be more vulnerable to the stress, anxiety, and cognitive decline associated with poor sleep than males [35]. Sex differences do exist in sleep-related disturbances and loss of sleep or insomnia is a more

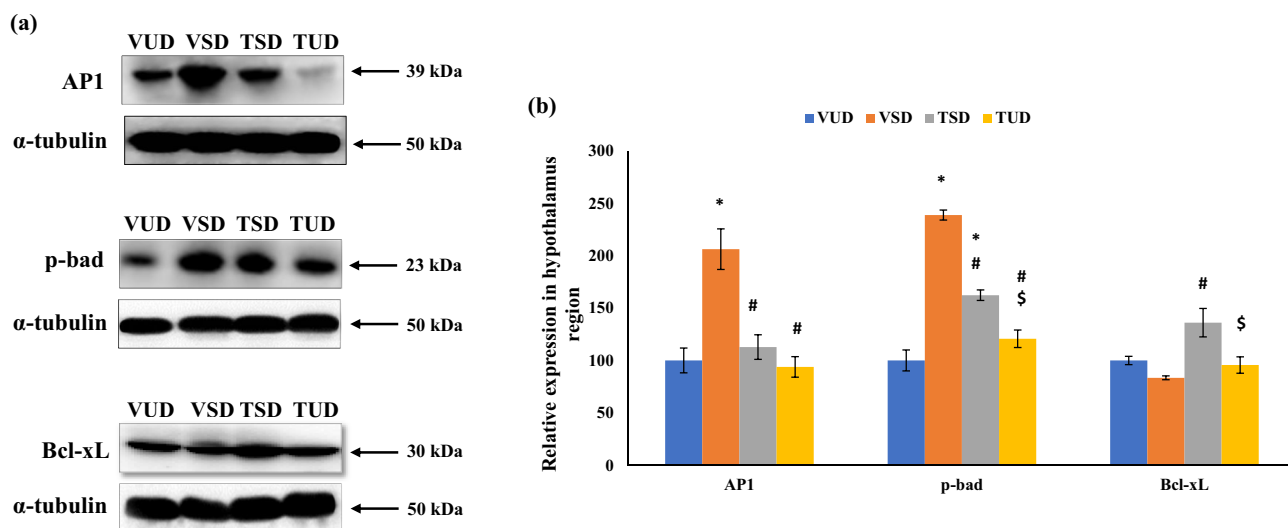


Fig. 6 B-TCE promoted BDNF-TrkB/PI3K-Akt pathway implicated in synaptic plasticity and cell survival: **a** representative western blot images of BDNF-TrkB signaling proteins: BDNF, TrkB, p-Akt, p-GSK3 β , and cyclinD1 in the hypothalamus region of VUD, VSD, TSD, and TUD rats ($n=6-7$ in each group). **b**, **c** Histograms rep-

resenting their quantitative densitometric analysis among different treatment groups are plotted as mean $\pm$ SEM. * $p \leq 0.05$ VUD versus VSD, TSD, TUD, # $p \leq 0.05$ VSD versus TSD and TUD, \$ $p < 0.05$ TSD versus TUD, Tukey's post hoc method after two-way ANOVA

prevalent problem in females [36, 37]. Women, as they age, become susceptible to the cumulative effects of menopause and often suffer from sleep disturbances, inflammation, and adiposity which are attributed to a compromised immuno-neuro-endocrine axis mainly regulated at the hypothalamic level. Serum cortisol and pro-inflammatory cytokines such as IL-6, IL-1 β , TNF- α , MCP-1 levels were elevated after SD in these animals which may be the outcome of HPA-axis activation in response to SD-induced stress. These observations are supported by clinical reports of acute sleep loss elevated HPA-axis functioning among young healthy adults in the subsequent day [38]. Psychological stress-induced HPA axis hyperactivation resulting in cortisol release has been found to increase pro-inflammatory cytokine secretion as a neuroendocrine homeostatic response to help prepare organism cope against the stressor [39, 40]. Further, this study provides first-ever preclinical evidence for the anxiolytic potential of B-TCE against acute SD. Individuals with lower sleep duration experience anxiety, depressive symptoms, cognitive dysfunctions, and reduced attention to various non-emotional stimuli [41]. Stressful signals are assessed by the neural network of core limbic system regions such as the amygdala, locus coeruleus, periaqueductal gray matter, and the hypothalamus which are responsible for mounting an adequate physiological and behavioral response. Hypothalamus being a key component of the HPA axis is specifically responsible for the metabolic and neuroendocrine responses to stress, which is inextricably linked to anxiety [42, 43].

Acute SD evoked an anxiogenic response in the VSD group animals as is evident from the EPM data of less time

spent in the open arm thus depicting their less exploratory activity on the maze. Similarly, VSD animals showed a fewer number of entries and crossings in the open arms as well as significantly lesser number of head dips compared to VUD rats. These changes were ameliorated in the TSD group pre-treated with B-TCE (Fig. 1). The anxiety-like behavior of VSD group animals may be attributed to HPA axis hyperactivity as evident from upregulation in serum cortisol levels which was suppressed by B-TCE pre-treatment (Fig. 2). SD elicited stress response by activation of HPA axis [38] may explain the current findings of an increase in cortisol secretion in the VSD group which is known to induce oxidative stress markers release thus causing anxiety and cognitive impairments. Further, consistent with this observation is the increase in protein expression of GFAP (a marker of reactive astrogliosis) in the hypothalamus and serum pro-inflammatory cytokines IL-6, TNF- α , IL-1 β , and MCP-1 secretion in SD animals which was abrogated by B-TCE supplementation (Fig. 3). The current data suggest that acute sleep loss resulted in HPA axis activation which in turn promoted pro-inflammatory cytokines secretion. The pro-inflammatory cytokines mediated activation of astrocyte signaling, in turn, promotes inflammatory activities in the CNS and causes impairments in neuronal functions affecting complex behaviors such as anxiety, cognition, appetite, and sleep [44]. Previous studies from our lab and others have reported that both acute and chronic forms of sleep deprivation activated microglia and astroglial cells in discrete brain regions [32, 45–48] thus supporting the current observations.

The ameliorative activity of B-TCE to suppress the changes in these markers is in line with another recent study which reported that aqueous extract of *T. cordifolia* suppressed MPTP-induced upregulation in the expression of GFAP and Iba1 [29]. Recently, our lab also reported the anti-neuroinflammatory activity of B-TCE against glutamate-mediated excitotoxicity using both in vivo and in vitro model systems [33, 34]. The increase in pro-inflammatory cytokines is also viewed as a neuroendocrine homeostatic response of an organism to cope with external stressors [39]. The current data thus provides compelling evidence that B-TCE exerts its anti-inflammatory activity simultaneously at both peripheral and CNS levels. The data also supports the conventional role of the hypothalamus in modulating SD-associated stress, anxiety, depression, and several metabolic disorders as sleep debt adversely impacts autonomic functions regulated at the level of the hypothalamus [49]. Stress-induced HPA axis hyperactivity results in a decline in monoamine neurotransmitters in both the peripheral and CNS as well as elevation in serum cortisol levels [50]. Reduced cortisol levels and anxiety in TSD animals may also be attributed to inhibition of brain monoamine oxidase (MAO-A and MAO-B) activity of *T. cordifolia* which has been reported to increase the concentration of monoamines such as serotonin, noradrenaline, and dopamine while decreasing the concentration of GABA in a mice model of depression [30]. Moreover, *T. cordifolia* has also been reported to enhance mitochondrial complex-1 activity and dopamine levels thus resulting in reduced oxidative stress [31].

Leptin and insulin level were reduced in the VSD group animals as compared to VUD animals which were augmented in B-TCE pre-treated animals (Fig. 2). The current study although reports the effects of only acute sleep deprivation but the decline in leptin levels caused by chronic curtailment of sleep time and the ensuing increase in appetite and weight gain have become the hallmark of modern society. One possible mechanism suggested for SD-associated decline in leptin levels is the increase in sympathetic nervous system activity due to sleep deprivation as evident from several pre-clinical and clinical studies reporting an increase in urine and plasma catecholamine levels after acute sleep loss [51, 52]. Therefore, reduced leptin secretion in VSD group animals may be the outcome of the inhibitory effect of increased sympathetic outflow [53]. Further SD-induced decline in leptin levels has been shown to result in a significant decrease in glucose tolerance, lower insulin release, and low-grade systemic inflammation [54]. The current data also corroborates the study by [55] which showed that SD reduced the concentration of anabolic hormones namely, testosterone, insulin, and leptin, and increased the levels of catabolic hormones such as noradrenaline, corticosterone, ACTH, and glucagon.

Keeping in view the role of different hypothalamic nuclei in the integration of stress-related signals from diverse brain regions to facilitate neuroendocrine response to stress [56], we further studied the expression of neuronal plasticity and stress response proteins such as PSA-NCAM, NCAM, nNOS, MAP-2, and HSP-70 in the hypothalamic region of the brain after acute SD. Expression of NCAM and its post-translationally modified form, PSA-NCAM as well as HSP-70 was upregulated in SD animals and this change was suppressed by B-TCE pre-treatment in TSD group animals. An increase in the expression of these plasticity and stress response markers which are hallmarks of synaptic remodeling may be an adaptive neuroprotective mechanism in VSD group animals against acute SD induced stress. NCAM and PSA-NCAM as key molecules of synaptic plasticity are associated with learning and memory consolidation and several previous studies have reported upregulation in their expression under adverse stressful conditions such as stress, anxiety, fear, juvenile, and social isolation [57, 58]. Further, it may be suggested that B-TCE pre-treatment of SD animals suppressed the acute stress-induced synaptic remodeling in order to maintain the structural and functional integrity of brain tissue. This is also evident from the anxiolytic effects observed in TSD group animals showing prevention of SD-induced changes in anxiety-like behavior. Forceful SD for 5 h has been reported to increase brain temperature as well as muscle tone during active wakefulness [59]. Another study [60] reported that animals exposed to hyperthermia prior to kainate-induced excitotoxic cell damage upregulated NCAM polysialylation in the hippocampus region. Findings of these previous studies and the current data further validate that upregulation of NCAM, PSA-NCAM, and HSP-70 by acute SD-induced hyperthermia may be a preconditioning-induced endogenous neuroprotective response. Besides its role as a marker of neuronal plasticity, PSA-NCAM along with BDNF signaling has been recently reported to mediate the function of NMDA receptor GluN2A subunit to abolish LPS induced inflammation-related depression [61]. The current findings of SD-induced anxiety-like behavior and underlying increase in pro-inflammatory markers along with upregulation in PSA-NCAM and NCAM expression in the hypothalamus region are supported by the previous study depicting the role of PSA-NCAM in inflammation.

NO plays a key role in CNS functions ranging from neural transmission to excitotoxicity [62]. In the sleep–wake cycle, inhibition of NO production has been reported to suppress spontaneous sleep while administration of NO donors promoted sleep [63]. These previous reports corroborate our current findings of nNOS suppression in the hypothalamus region thus suggesting that one possible mechanism by which acute SD disrupts homeostatic regulation of the sleep–wake cycle is by inhibiting NO production, a key signaling molecule in the modulation of sleep and memory consolidation [64]. MAP-2

plays important role in synaptic plasticity by modulation of dendritic remodeling [65]. Downregulation in the expression of MAP-2 may indicate the abnormal assembly of microtubules and cytoskeletal proteins in response to acute SD but was augmented by B-TCE supplementation (Fig. 4).

Further to explore whether BDNF, a key neurotrophic signaling molecule is impacted by acute sleep loss in the hypothalamus and is targeted by B-TCE to maintain neuronal plasticity, expression of BDNF, and its receptor TrkB was also studied. Their expression was downregulated in the VSD group which was suppressed in TSD animals thus suggesting that B-TCE supplementation promoted cell survival and synaptic plasticity (Fig. 5). Reduction in BDNF and TrkB expression has been associated with anxiety-like behavior in different animal models [66, 67]. BDNF on binding to TrkB further leads to activation of various downstream cell survival signaling pathways such as PI3K/Akt which are imperative in mediating neuroprotection in a wide variety of ways [68]. Once activated, p-Akt promotes cell survival and suppresses neuronal apoptosis by enhancing GSK-3 β phosphorylation at ser9 which in turn leads to accumulation of cyclinD1 in the nucleus resulting in cell cycle progression [69]. B-TCE pre-treatment prior to acute SD enhanced the expression of p-Akt, p-GSK3 β , and cyclinD1 which was otherwise downregulated by SD in the VSD group animals (Fig. 5). The data, therefore suggest that the active ingredients present in B-TCE promoted neuronal survival and plasticity by targeting the BDNF-TrkB pathway and downstream Akt/GSK3 β /Cyclin D1 molecules. Further, the anti-apoptotic activity of B-TCE is also supported by downregulated expression of pro-apoptotic proteins p-bad and AP-1 and enhanced level of anti-apoptotic protein, Bcl-xL in the TSD group as compared to VSD animals (Fig. 6). Previous LCMS-based studies from our lab have reported the presence of oblongine, palmatine, tetrahydropalmatine, magnoflorine, norcoclaurine, tinocordiside, 11-hydroxy mustakone, and cordifolioside A in B-TCE [33]. Similarly, few other studies have also reported the presence of alkaloids and glycosides in the n-Butanol fraction of *T. cordifolia* stem extract [33, 70]. Different alkaloids and phenols from *T. cordifolia* are known to possess antioxidant properties and are suggested as key players to prevent neurodegeneration by suppressing ROS formation [71–73]. Future studies may look into the role of these active components of B-TCE for the management of SD-induced anxiety and cognitive impairments.

Conclusion

The current findings have identified multiple targets of B-TCE activity such as inflammation, synaptic plasticity, apoptosis, and the BDNF/TrkB pathway of cell survival.

It may be suggested that a combination of diverse ingredients present in B-TCE rich in antioxidants may have contributed in ameliorating anxiety and cognitive impairments associated with acute SD. Further, a detailed study is required in the future to explore the role of active ingredients in the B-TCE that may be responsible for ameliorating SD-induced anxiety-like behavior. Further, more rigorous studies using specific pathway inhibitors will help delineate the molecular targets of B-TCE in providing neuroprotection.

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Author Contributions PB and GK designed the study. PB carried out the experiments. PB and GK analyzed the data and wrote the manuscript. GK provided the infrastructure to carry out the work. Both the authors have read and approved the final manuscript.

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Data Availability All data generated or analysed during this study are included in this published article.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical Approval All animal experimental protocols were approved by the Institutional Animal Ethical Committee (IAEC) of Guru Nanak Dev University, Amritsar, Punjab, India (Reference no: 226/CPC-SEA/2019/10). Protocols were performed according to the guidelines of 'Animal care and use' laid down by IAEC.

Consent to Participate Not applicable.

Consent to Publish Not applicable.

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